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The challenges and opportunities of Artificial Intelligence in education



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Artificial Intelligence (AI) is producing new teaching and learning solutions that are currently being tested globally. These solutions require advanced infrastructures and an

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These are some of the questions explored in a [Working Paper entitled 'Artificial Intelligence in Education: Challenges and Opportunities for Sustainable Development'](#) presented by UNESCO and ProFuturo at [Mobile Learning Week 2019](#). It features cases studies on how AI technology is helping education systems use data to improve educational equity and quality.

Concrete examples from countries such as China, Brazil and South Africa are examined on AI's contribution to learning outcomes, access to education and teacher support. Case studies from countries including the United Arab Emirates, Bhutan and Chile are presented on how AI is helping with data analytics in education management.

The Paper also explores the curriculum and standards dimension of AI, with examples from the European Union, Singapore and the Republic of Korea on how learners and teachers are preparing for an AI-saturated world.

Beyond the opportunities, the Paper also addresses the challenges and policy implications of introducing AI in education and preparing students for an AI-powered future. The challenges presented revolve around:

- **Developing a comprehensive view of public policy on AI for sustainable development:** The complexity of the technological conditions needed to advance in this field require the alignment of multiple factors and institutions. Public policies have to work in partnership at international and national levels to create an ecosystem of AI that

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technological infrastructure must be faced to establish the basic conditions for implementing new strategies that take advantage of AI to improve learning.

- **Preparing teachers for an AI-powered education:** Teachers must learn new digital skills to use AI in a pedagogical and meaningful way and AI developers must learn how teachers work and create solutions that are sustainable in real-life environments.
- **Developing quality and inclusive data systems:** If the world is headed towards the datafication of education, the quality of data should be the main chief concern. It's essential to develop state capabilities to improve data collection and systematization. AI developments should be an opportunity to increase the importance of data in educational system management.
- **Enhancing research on AI in education:** While it can be reasonably expected that research on AI in education will increase in the coming years, it is nevertheless worth recalling the difficulties that the education sector has had in taking stock of educational research in a significant way both for practice and policy-making.
- **Dealing with ethics and transparency in data collection, use and dissemination:** AI opens many ethical concerns regarding access to education system, recommendations to individual students, personal data concentration, liability, impact on work, data privacy and ownership of data feeding algorithms. AI regulation will require public

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a unique opportunity to explore together the opportunities and threats of AI in all areas of education.

- [Download the working paper](#)

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